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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

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1. Team & Project Overview

Team Name	NullTeam
Selected Domain	1
Selected SDG Track (1–17)	4,8
Project / Solution Name	jAIX (Job Artificial Intelligence Expert)
Name (Leader)	Muhammad Haqim bin Mohd Nazry
Email & Phone (Leader)	haqimnazry@gmail.com & +60 16-240 4652
University Affiliation(s)	Universiti Malaya
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

AI is rapidly reshaping the labor market, displacing routine, rules-based roles while creating demand for new AI-adjacent and sustainability-oriented occupations. Workers who cannot anticipate these shifts risk being left behind. Working adults in administrative and clerical roles, fresh graduates entering a volatile job market, and HR teams planning workforce transformation are all affected. There is no lightweight, scenario-aware tool that helps individuals and HR teams map current skills to resilient, growing roles.

This matters because unmanaged displacement deepens economic inequality and skills mismatch. Our dataset shows declining, high-disruption roles like Data Entry Clerk and Routine Reporting Assistant alongside growing, lower-disruption roles like AI Governance Analyst, Prompt Engineer, Renewable Energy Analyst. Roughly 56% of roles face meaningful disruption exposure while only about 26% of workers are on clearly growing, future-ready pathways. Without intervention, this gap will widen, leaving vulnerable workers without actionable guidance on where to reskill or which Sustainable Development Goals their career transitions can support.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

jAIX (Job Artificial Intelligence Expert) is a dynamic web platform that guides users from current skills to AI-era readiness through a 4-step journey: Intake, Trend Radar, Role Matching, and Reskilling Pathways.

Intake: Users input their current skills.

Trend Radar: Surfaces role growth outlooks and disruption risks ingested from live data feeds and job trends.

Role Matching: Applies explainable, rule-based skill-fit scoring to rank compatible roles and break down missing skill gaps.

Reskilling Pathways: Generates ordered learning roadmaps using free online courses and accredited programs.

The platform models multiple disruption scenarios (Moderate vs. Rapid/Fast AI Disruption) so users can stress-test recommendations under different labor-market assumptions. Crucially, every skill and career path is mapped directly to UN Sustainable Development Goals (SDGs 4, 8, 9).

Core matching, trend filtering, and SDG mapping are implemented as local, rule-based logic in TypeScript, no third-party AI model is required for core functionality, making results transparent, auditable, and free of automated hiring bias. A serverless RunPod GPU endpoint running Qwen2.5-7B-Instruct extracts resume text into structured JSON for deterministic role matching.

Tools and skills from SDC Phase 1 workshops, GenAI APIs, database architecture, CSV data ingestion, and cloud hosting on Replit, are leveraged to build a functional, deployed MVP.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: React + Vite, Tailwind CSS, Radix UI, Framer Motion

Backend: Node.js, Express 5, Pino (logging)

Data / CSV Parsing: None

Validation & API Contract: Zod, Orval, OpenAPI spec

Database / ORM: File-backed SQLite database hosted on Replit

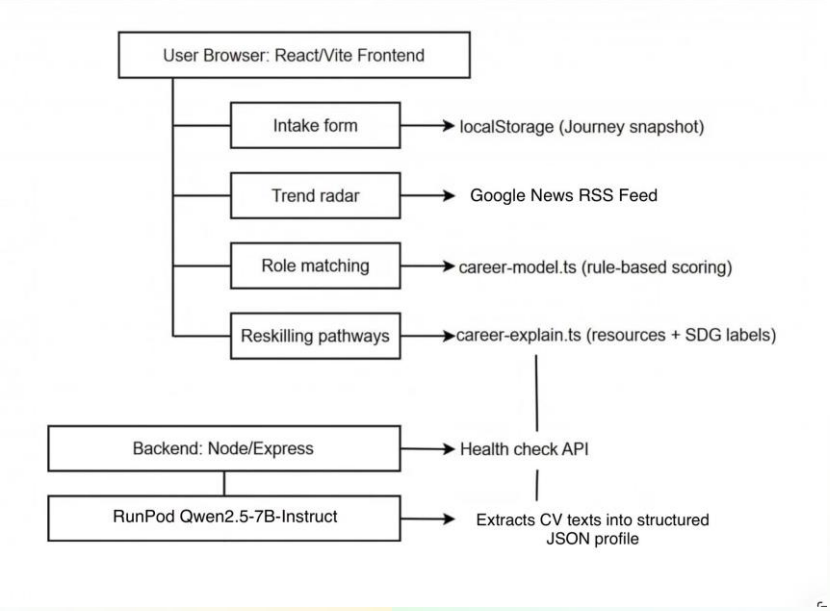
Client Storage: Browser localStorage with Zod-validated journey snapshots

Package / Runtime: pnpm workspaces, TypeScript 5.9, Node.js 24

Cloud / Hosting & Dev Environment: Replit (cloud IDE, automated workspaces, deployment hosting)

GenAI / AI Tooling: RunPod GPU endpoint (Qwen2.5-7B-Instruct)

External APIs: Google News RSS (live trend data feed), Unsplash CDN (fallback placeholder images)



3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We applied design thinking by empathizing with working adults, fresh graduates, and HR planners, defining the core problem of AI-driven job displacement; ideating a four-step user journey, prototyping rapidly in Replit, and validating against live RSS job trend data. The modular monorepo (separate frontend, backend, and shared schema packages) ensures technical viability, while our file-backed SQLite database architecture ensures lightweight, deployable functionality for Phase 2. User feedback shaped the Moderate vs. Fast scenario toggle and SDG mapping, keeping the solution human-centered, explainable, and ready for real deployment by Phase 2. Iterative testing in Replit's cloud environment confirmed cross-device functionality and offline resilience.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 4.4: Substantially increase the number of youth and adults with relevant skills for employment and entrepreneurship. Target 8.5: Achieve full and productive employment and decent work for all women and men.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	600,000+ Malaysian workers require urgent reskilling; 74% lack formal AI training while 56% use unguided AI tools.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Average time to generate a verified, SDG-mapped reskilling pathway and percentage of recommended courses verified against accredited frameworks
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Guide target learners onto verified reskilling pathways with 100% SDG mapping transparency

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
1. React	**Code**	**MIT**
2. Vite	**Code**	**MIT**
3. Tailwind CSS	**Code**	**MIT**
4. Radix UI	**Code**	**MIT**
5. Framer Motion	**Code**	**MIT**
6. Express	**Code**	**MIT**
7. Zod	**Code**	**MIT**
8. Orval/OpenAPI	**Code**	**MIT**
9. PDF.js	**Code**	**Apache 2.0 / MIT**
10. Replit	**Cloud / Dev platform**	**Proprietary**
11. Qwen2.5-7B-Instruct	**Model**	**Apache 2.0**

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



Muhammad Haqim bin Mohd Nazry

24 October 2026